

CHEM.

Use of Chemistry in day to day life

• Use of phenolphthalein - Detection of nature (acid / base) of waste water.

* Used in anticorruption Dept.

* Used in soil testing

• Use of H_2SO_4 - Used in manufacturing of ammonium sulphate (fertilizer)

* It is used in refining petroleum.

* It is used in production of various salt.

* It is very important acid for chemical detection (testing).

* It is very widely used in synthesis of detergents.

11/12/2

Q) What is alkali? Give eq of it.

A All soluble bases are called alkali.
eg NaOH , KOH , RbOH , LiOH etc.

Q) What are weak bases? Give eq.

A The Hydroxide of alkaline metals are weak bases

eg $\text{Mg}(\text{OH})_2$, $\text{Be}(\text{OH})_2$

$\text{Ca}(\text{OH})_2$

$\text{Sr}(\text{OH})_2$

$\text{Ba}(\text{OH})_2$

~~Ra~~

Q) What is anhydrous salt? Give eq.

A Salt without water molecules is called anhydrous salt.

eg - CuSO_4 , ZnSO_4 , FeSO_4

Q) What are hydrated salt? Give eq.

A Salt crystal containing water molecule is called Hydrated salt. eg. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

HIST.

at
Iron pillar ~~near~~ Mehrauli (Delhi),
is over 1600 yrs old and still stands
without rusting and is an eg-
of its metallurgical skill of
ancient India.

11/12/25

B,

library source - Vishnu Khanda
which specifies the key regions
of Empire - It ruled over
Angara (the middle gangetic basin),
Prayag, Saketa (Ayodhya) and
Magadha (Bihar)

Q) Briefly discuss the achievements of
Chandragupta I.

A. The first powerful ruler of Gupta
Empire of Chandragupta I who
played a crucial role in the
early expansion of Gupta empire.

• He is remembered for his
coins for eg. gold coin featuring
King Chandragupta I with his
~~wife~~ queen Kumara Devi on
the reverse side a seated goddess

identified as Lakshmi was found.

- He was also famous for his strategic alliances which helped him consolidate his power and led to the foundation of strong empire.

- He also took the title of 'Maharaja' (Chakravartin) means King of the Kings.

Q) Briefly discuss the military achievements of Samudragupta.

Source of Information - Prayaga Prashasti or Allahabad pillar inscription was composed by his court poet Harisena in Sanskrit language.

- In this inscription Harisena explained that the king's ambition was to be 'Dharmachakravartin' or 'to unify the earth'. In this

context he fought many wars.

Military achievement

- He defeated many kings, taking over the kingdoms and expanding his empire.
- Many defeated kings offered tribute to Samudragupta while others fearing his might submitted without protests.

Qualities / Characteristics

- He supported art, learning, trade and making his kingdom rich and successful.
- He himself is ~~not~~ Portrayed as a Veena player in one of the coins he minted.

MATH

~~Length and breadth~~

~~It is~~

- Q) One side of a parallelogram is 14 cm. Its distance from the opposite side is 16.5 cm. Find the area of the parallelogram.

$$b = 14 \text{ cm} ; h = 16.5 \text{ cm}$$

$$\text{Area of the } \parallel \text{gm} = b \times h$$

$$= 14 \times 16.5 \text{ cm}^2$$

$$= \frac{7}{14} \times \frac{165}{10} \times 2 \text{ cm}^2$$

$$= 231 \text{ cm}^2$$

- Q) Find the area of the rhombus, the lengths of whose diagonals are 36 cm and 22.5 cm.

$$d_1 = 36 \text{ cm}$$

$$d_2 = 22.5 \text{ cm}$$

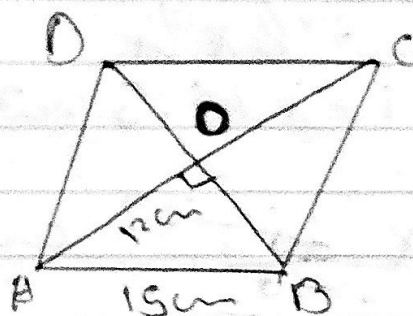
Area of a rhombus

$$= \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 36 \times 22.5 \text{ cm}^2$$

$$= \frac{1}{2} \times 18 \times \frac{225}{10} \text{ cm}^2$$

$$= 405 \text{ cm}^2$$

- * Q) Find the area of a Rhombus in which each side is 15 cm long and one of whose diagonals is 24 cm.



Let ABCD be the rhombus in which $AB = 15 \text{ cm}$ and $AC = 24 \text{ cm}$

$\therefore$ We know that the diagonals of a rhombus bisect each other at right angles.

$$\therefore OA = \frac{1}{2} \times AC = \left(\frac{1}{2} \times 24 \right) \text{ cm} = 12 \text{ cm}$$

$$\angle AOB = 90^\circ$$

$$OB = \sqrt{AB^2 - OA^2} = \sqrt{(15)^2 - (12)^2} \text{ cm}$$

$$= \sqrt{225 - 144} \text{ cm} = \sqrt{81} \text{ cm} = 9 \text{ cm}$$

$$\therefore BD = (2 \times OB) = (2 \times 9) \text{ cm} = 18 \text{ cm}$$

$$\begin{aligned} \text{Area of rhombus ABCD} &= \left(\frac{1}{2} \times AC \times BD \right) \text{ sq. unit} \\ &= \left(\frac{1}{2} \times 24 \times 18 \right) \text{ cm}^2 = \boxed{216 \text{ cm}^2} \end{aligned}$$

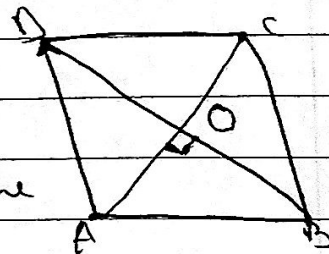
* Q) If the area of a rhombus is 96 cm^2 and one of its diagonals is 16 cm , find the perimeter.

Let the other diagonal of the given rhombus be $u \text{ cm}$.

$$\text{Then, } \frac{1}{2} \times 16 \times u = 96 \Rightarrow 8u = 96$$

$$\Rightarrow u = 12$$

Thus, the length of the other diagonal is 12 cm



Let us consider a rhombus ABCD in which diagonal AC = 12 cm and diagonal BD = 16 cm. We know that the diagonals of a rhombus bisect each other at right angles.

$$\therefore OA = \frac{1}{2} AC = 6 \text{ cm}, OB = \frac{1}{2} BD = 8 \text{ cm}$$

$$\angle AOB = 90^\circ$$

$$\therefore AB^2 = OA^2 + OB^2 = (6)^2 + (8)^2 = (36 + 64) = 100$$

$$\Rightarrow AB = \sqrt{100} \text{ cm} = 10 \text{ cm}$$

Thus, each side of the given rhombus is 10 cm.

$$\therefore \text{Hence the perimeter is } (10 \times 4) \text{ cm} = \boxed{40 \text{ cm}}$$

EX-20C

- 1) Find the area of a llgm with base 32 cm and height 16.5 cm.

$$\begin{aligned}
 b &= 32 \text{ cm}; h = 16.5 \text{ cm} \\
 \text{area of llgm} &= b \times h \\
 &= 32 \times 16.5 \text{ cm}^2 \\
 &= \cancel{32}^{\cancel{16}} \times \cancel{165}^{\cancel{10}} \text{ cm}^2 \\
 &= 528 \text{ cm}^2
 \end{aligned}$$

- 2) The base of a llgm measures 1 m 60 cm and its height is 75 cm. Find its area in m².

$$b = 1.60 \text{ m}; h = 0.75 \text{ m}$$

$$\begin{aligned}
 \text{Area of llgm} &= (b \times h) \text{ m}^2 \\
 &= \left(\begin{array}{r} \cancel{160}^{\cancel{100}} \times \cancel{75}^{\cancel{100}} \\ \hline \cancel{5}^{\cancel{10}} \end{array} \right) \text{ m}^2 = \frac{12}{10} \text{ m}^2 \\
 &= 1.2 \text{ m}^2
 \end{aligned}$$

3) In a llgm it is being given that
base = 14 dm and height = 6.5 dm
Find its area in

(i) cm^2

$$b = 140 \text{ cm}$$

$$h = 65 \text{ cm}$$

$$\text{Area of llgm} = (b \times h) \text{ cm}^2$$

$$= (140 \times 65) \text{ cm}^2$$

$$= 9100 \text{ cm}^2$$

(ii) $\text{m}^2 = 0.91 \text{ m}^2$

SANSKRIT



अभ्यासाः



सम्प्रति लेखनीयम्

21st
Century
Skills

Conceptual
Understanding

1. प्रश्नानाम् उत्तराणि संस्कृतेन लिखत।

(प्रश्नों के उत्तर संस्कृत में लिखिए। Answer the questions in Sanskrit.)

(क) प्रातःकाले उत्थाय नरः किं कुर्यात्?

प्रातःकाले उत्थाय नरः स्निग्धं भोजनं।

(ख) किं व्यायामः वृद्धेभ्यः अपि लाभप्रदः?

आयुः, व्यायामः वृद्धेभ्यः अपि लाभप्रदः।

(ग) किं नरः मुहुर्मुहुः खादेत्?

नरः, नरः मुहुर्मुहुः न खादेत्।

(घ) जीवने कः सर्वाधिकं महत्वपूर्णः भवति?

जीवने सत्यः सर्वाधिकः महत्वपूर्णः भवति।

2. मञ्जूषायां दत्तैः पदैः रिक्तस्थानानि पूरयत।

(मञ्जूषा में दिए शब्दों से रिक्त स्थानों की पूर्ति कीजिए। Fill in the blanks with the words given in the box.)

मञ्जूषा - वृद्धेभ्यः, खादेत्, स्मरेत्, उत्तिष्ठेत्, कुर्युः

(क) किं नरः पुनः पुनः खादेत् ?

(ख) यूयं ध्यानेन स्वपाठं स्मरेत् ।

(ग) व्यायामः वृद्धेभ्यः अपि लाभप्रदः।

(घ) यूयं सदा सूर्योदयात् पूर्वम् उत्तिष्ठेत् ।

(ङ) छात्राः सर्वाणि कार्याणि समये कुर्युः ।

3. 'खाद्' धातोः विधिलिङ्लकारे रूपाणि लिखत।

('खाद्' धातु के विधिलिङ् लकार में रूप लिखिए। Conjugate the roots 'खाद्' in विधिलिङ् लकार)

धातु 'खाद्' (खाना) विधिलिङ् लकार

पुरुषः	एकवचनम्	द्विवचनम्	बहुवचनम्
प्रथमः पुरुषः	- खादेत्	- खादेताम्	- खादयुः
मध्यमः पुरुषः	- खादे	- खादेताम्	- खादे
उत्तमः पुरुषः	- खादेताम्	- खादेताम्	- खादेताम्

4. शुद्धम् उत्तरं (✓) इति चिह्नेन चिह्नीकुरुत।

(शुद्ध उत्तर पर सही का निशान (✓) लगाइए। Tick the correct equivalent.)

- (क) खाद् विधिलिङ् लकार प्रथम पुं द्विवचन - खादेतम् ☐ खादेताम् ☒ खादेयुः ☐ खादेः
- (ख) पा-पिब विधिलिङ् लकार मध्यम पुं एकवचन - पिबेत् ☒ पिबेः ☒ पिब ☐ पिबैत्
- (ग) गम्-गच्छ विधिलिङ् लकार उत्तम पुं बहुवचन - गच्छेयम् ☐ गच्छेव ☐ गच्छेम ☒ गच्छेः
- (घ) पठ् विधिलिङ् लकार उत्तम पुं द्विवचन - पठेम ☒ पठेव ☐ पठेतम् ☐ पठेत
- (ङ) उत्-स्था विधिलिङ् लकार उत्तम पुं एकवचन - उत्तिष्ठेत् ☐ उत्तिष्ठेः ☐ उत्तिष्ठेयम् ☒ उत्तिष्ठेम

5. संस्कृतेन अनुवादं कुरुत।

(संस्कृत में अनुवाद कीजिए। Translate into Sanskrit.)

- (क) हमें प्रातःकाल भ्रमण के लिए जाना चाहिए। वयं प्रातःकाले भ्रमणाय गच्छामः।
- (ख) छात्रों के जीवन में समय बहुत महत्वपूर्ण है। छात्राणां जीवने समयः महत्त्वपूर्णः।
- (ग) हमें सब काम समय पर करना चाहिए। वयं सर्वानि कार्याणि समये कुर्यामः।
- (घ) वृद्धों को भी यथाशक्ति व्यायाम करना चाहिए। वृद्धाः यथाशक्त्या व्यायामं कुर्यान्।
- (ङ) मनुष्य को सूर्योदय से पहले उठना चाहिए। नरः सूर्योदयं दृष्ट्वा उतिष्ठेत्।

6. उचितेन पदेन रिक्तस्थानपूर्ति कुरुत।

(उचित पद से रिक्त स्थानों की पूर्ति कीजिए। Fill in the blanks with correct word.)

- (क) यूयं प्रातः भ्रमणाय । (गच्छेत / गच्छेतम् / गच्छेः)
- (ख) सर्वे समये कार्याणि । (कुर्यात् / कुर्याताम् / कुर्युः)
- (ग) सायंकाले यूयं क्रीडाक्षेत्रे । (क्रीडेयुः / क्रीडेत् / क्रीडेम)
- (घ) वयं स्वास्थ्यस्य रक्षां । (कुर्याम / कुर्यात् / कुर्युः)
- (ङ) नरः पुनः पुनः न । (खादेत् / खादेः / खादेयम्)

DO CHECK

The H.W

and NOTICES